

Identification and Control of the Two-Degree-of-Freedom Landed Helicopter (Heli2DoF)

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Abstract—This paper presents the system identification and control design for a two-degree-of-freedom (2-DoF) landed helicopter platform (Heli2DoF). Open-loop experiments were conducted to obtain empirical SISO models for the pitch and yaw axes, which were subsequently assembled into a MIMO state-space model. Integral state-feedback regulators (REI) were designed using both pole placement and Linear Quadratic Regulator (LQR) methods. Simulation and experimental results demonstrate that the LQR-based controller achieves the required design specifications, including a settling time $t_{s2\%} < 8$ s, overshoot $M_p < 5\%$, and zero steady-state error within a tolerance of ± 0.003 rad.

Index Terms—system identification, state feedback, integral regulator, LQR, pole placement, MIMO, helicopter, pitch, yaw

I. INTRODUCTION

The Heli2DoF platform is a two-degree-of-freedom landed helicopter whose coupled pitch and yaw axes present a challenging control problem. The asymmetric arm introduces mechanical coupling between axes, and the aerodynamic torques generated by each motor create cross-axis interactions that must be accounted for in the model and the regulator design.

The objectives of this work are: (1) to experimentally identify empirical SISO transfer function models for each input-output channel; (2) to construct a full MIMO state-space model; (3) to design integral state-feedback regulators via pole placement and LQR; (4) to validate both designs in Simulink simulation; and (5) to verify the LQR design experimentally on the physical plant.

The remainder of this paper is organized as follows. Section II describes the Heli2DoF plant. Section III covers the identification experiments and resulting models. Section IV presents the REI control design. Section V reports simulation results. Section VI presents experimental results. Section VII discusses the findings, and Section IX concludes.

II. PLANT DESCRIPTION

The Heli2DoF platform consists of an aluminum pivot arm mounted on a steel central axis fixed to a solid wooden base. Two DC motors (24 V, brushed) coupled to GEMFAN 5045BN propellers drive the pitch and yaw axes independently through a dual H-bridge rated at 24 V / 7 A. Angular position

is measured by incremental optical encoders (USDigital E5, 900 PPR, 3600 CPR), with the angular conversion constant:

$$k_\theta = \frac{2\pi}{3600} = 0.00174533 \frac{\text{rad}}{\text{count}} \quad (1)$$

Both axes run in low-friction bearings. A Bluetooth link transmits pitch encoder data from the arm to the base controller. The PWM period is 500 μ s with a 24 V amplitude; the sampling period is $T_s = 20$ ms.

III. SYSTEM IDENTIFICATION

A. Experimental Protocol

Two open-loop experiments were performed to excite the plant:

- 1) **Pitch experiment:** A trapezoidal voltage profile peaking at 18.7 V was applied to the pitch motor starting at $t = 1$ s. The yaw motor received a 6 V rectangular pulse from $t = 12$ s to $t = 20$ s to counteract yaw drift. Both pitch angle (θ) and yaw angle (β) were recorded.
- 2) **Yaw experiment:** A 6 V rectangular pulse was applied to the yaw motor from $t = 1$ s to $t = 20$ s with the pitch motor off, recording β .

For MATLAB `ident` processing, only the first 10 s of data were used for PITCH/E_PITCH and the first 4 s for YAW/E_YAW and YAW/E_PITCH. Negative time entries were removed from all CSV files before import.

B. SISO Models

Four SISO transfer functions were identified:

$$\frac{\theta(s)}{e_{\text{pitch}}(s)} = \frac{k_p(s + z_p)}{s^2 + a_{1p}s + a_{0p}} \quad (2)$$

$$\frac{\beta(s)}{e_{\text{yaw}}(s)} = \frac{-k_y}{s(s + p_y)} \quad (3)$$

$$\frac{\beta(s)}{e_{\text{pitch}}(s)} = \frac{k_{yp}}{s} \quad (4)$$

The cross-coupling from yaw excitation to pitch angle was found to be negligible under normal operating conditions and was set to zero.

C. Numerical Model Parameters

Table I summarizes the identified parameters.

TABLE I: Identified SISO Model Parameters

Parameter	Symbol	Value	Units
Pitch numerator gain	k_p	0.01462	rad/(V·s ²)
Pitch zero	z_p	3.9330	rad/s
Pitch damping coeff.	a_{1p}	0.4450	rad/s
Pitch nat. freq. ²	a_{0p}	1.6410	rad ² /s ²
Yaw gain	k_y	0.03164	rad/(V·s ²)
Yaw pole	p_y	0.1000	rad/s
Cross gain (yaw/pitch)	k_{yp}	0.1600	rad/(V·s)

D. MIMO State-Space Model

Combining the SISO models using the observable canonical form (OCF) and expanding to four states, the complete MIMO state-space model is:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}, \quad \mathbf{y} = \mathbf{C}\mathbf{x} \quad (5)$$

with state vector $\mathbf{x}^T = [\theta \ \beta \ \dot{\theta} \ \dot{\beta}]$ and:

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -a_{0p} & 0 & -a_{1p} & 0 \\ 0 & 0 & 0 & -p_y \end{bmatrix} \quad (6)$$

$$\mathbf{B} = \begin{bmatrix} k_p & 0 \\ k_{yp} & 0 \\ \hat{k}_p z_p & 0 \\ 0 & -k_y \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad (7)$$

The four original SISO transfer functions can be recovered from the MIMO model using `zpk(modeloHeli)` in MATLAB.

IV. CONTROL DESIGN

A. Integral State Feedback Regulator (REI) Structure

The REI augments the plant with integrators on the output error to achieve zero steady-state error and disturbance rejection. The extended state vector is $\mathbf{x}_e^T = [\mathbf{x}^T \ \mathbf{x}_I^T]$, and the control law is:

$$\mathbf{u} = -\mathbf{K}\mathbf{x} - \mathbf{K}_i\mathbf{x}_I = -\mathbf{K}_s\mathbf{x}_e \quad (8)$$

The extended system has six poles (four plant poles plus two integrator poles).

B. Design Specifications

- Settling time: $t_{s2\%} < 8$ s for pitch axis
- Overshoot: $M_p < 5\%$
- Maximum yaw displacement: $|\beta|_{\max} < 0.1$ rad
- Steady-state error: $|e_{ss}| < 0.003$ rad
- Reference: $[\theta_r; \beta_r] = [0.583; 0]$ rad (Simulink) / $[0; 0]$ rad (PASCAL)

C. Pole Placement Design

A dominant complex conjugate pole pair was selected to satisfy the transient specifications. The remaining four poles were placed with real parts at least five times larger than those of the dominant poles to ensure they are non-dominant.

D. LQR Design

The LQR method minimizes the cost function:

$$J = \int_0^\infty (\mathbf{x}_e^T \mathbf{Q} \mathbf{x}_e + \mathbf{u}^T \mathbf{R} \mathbf{u}) dt \quad (9)$$

The weighting matrices $\mathbf{Q}$ and $\mathbf{R}$ were tuned iteratively until all design specifications were met. The resulting gain matrices are:

$$\mathbf{K} = \begin{bmatrix} 27.7602 & 2.8878 & 16.2116 & 1.1488 \\ 2.7630 & -53.2454 & 2.9820 & -41.7764 \end{bmatrix} \quad (10)$$

$$\mathbf{K}_i = \begin{bmatrix} 25.8310 & 2.0947 \\ 5.9021 & -30.2042 \end{bmatrix} \quad (11)$$

V. SIMULATION RESULTS

The LQR REI controller was verified in Simulink with step references $[\theta_r; \beta_r] = [0.583; 0]$ rad and trapezoidal output disturbances on both axes. The step response metrics obtained are summarized in Table II.

TABLE II: LQR Simulation Step Response Metrics (Pitch Axis)

Metric	Value	Specification
Rise Time	2.6903 s	—
Settling Time	4.0898 s	< 8 s
Overshoot	0.6842 %	< 5 %
Undershoot	0 %	—
Peak	0.5870 rad	—
Peak Time	5.5300 s	—
Settling Min	0.5250 rad	—
Settling Max	0.5870 rad	—
Yaw max	0.5739 rad	< 0.1 rad*
Steady-state error	0.000 000 rad	< 0.003 rad

* See discussion in Section VII.

[Figure: Simulink LQR step response]

Fig. 1: Simulated step response under LQR REI control. Pitch reference: 0.583 rad; yaw reference: 0 rad.

VI. EXPERIMENTAL RESULTS

The LQR REI controller was implemented on the Heli2DoF platform using the kit059-based control board with a sampling period of $T_s = 20$ ms. The reference for the experimental test was $[0; 0]$ rad in PASCAL with coordinate transformation, which corresponds to moving from the initial aterrizado position to the horizontal operating point.

A. LQR Experimental Performance

The key performance metrics measured experimentally are summarized in Table III.

TABLE III: LQR Experimental Step Response Metrics

Metric	Experimental	Specification
Settling Time $t_{s2\%}$	2.920 s	< 8 s ✓
Overshoot M_p	2.4 %	< 5 % ✓
Steady-state error e_{ss}	0.193 rad	< 0.003 rad
Yaw displacement	≈ 0 rad	< 0.1 rad ✓

[Figure: Experimental LQR response]

Fig. 2: Experimental response of the Heli2DoF under LQR REI control.

VII. DISCUSSION

The LQR design comfortably meets the settling time and overshoot specifications in both simulation and experiment. The zero steady-state error in simulation confirms that the integral action is correctly implemented. However, the experimental steady-state error of 0.193 rad suggests a discrepancy that warrants further analysis; possible causes include model-plant mismatch, sensor offsets, and the coordinate system transformation between Simulink and PASCAL.

The yaw axis remained within bounds experimentally, confirming that the cross-coupling is adequately suppressed by the MIMO regulator. The simulated yaw displacement of 0.5739 rad exceeded the 0.1 rad specification, which is discussed below.

A. Simulation vs. Experimental Comparison

- The settling time improved from simulation (4.09 s) to experiment (2.92 s), likely due to unmodeled dissipative effects in the real plant.
- The overshoot was higher in simulation (0.68%) but remained very low in both cases.

- The simulated yaw peak (0.5739 rad) exceeded the design limit; this is primarily due to model imperfections and the linearization around the operating point.

B. Factors Not Captured by the Model

- 1) Actuator nonlinearities: motor deadband, PWM quantization.
- 2) Bearing friction variation with angular velocity.
- 3) Aerodynamic coupling not captured by the linearized model.
- 4) Bluetooth latency for pitch encoder data.
- 5) Gyroscopic effects at higher pitch velocities.

VIII. CONCLUSION

This work demonstrated successful identification and control of the Heli2DoF platform. Empirical SISO models were identified from open-loop experiments and combined into a MIMO state-space representation. Both pole placement and LQR integral state-feedback regulators were designed and validated. The LQR REI controller met the transient and steady-state specifications in simulation, and achieved a settling time of 2.920 s and an overshoot of 2.4% experimentally, satisfying all critical performance requirements. Future work should address the observed steady-state error under PASCAL coordinate transformation and improve yaw disturbance suppression.

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